Aquifer Memory: How France’s Groundwater Recovered from the 2022 Drought

Fissured aquifers recovered fastest regardless of drought severity; recovery time governed primarily by drought depth, not aquifer medium alone

Geodel · GroundwatCH Programme · November 2026

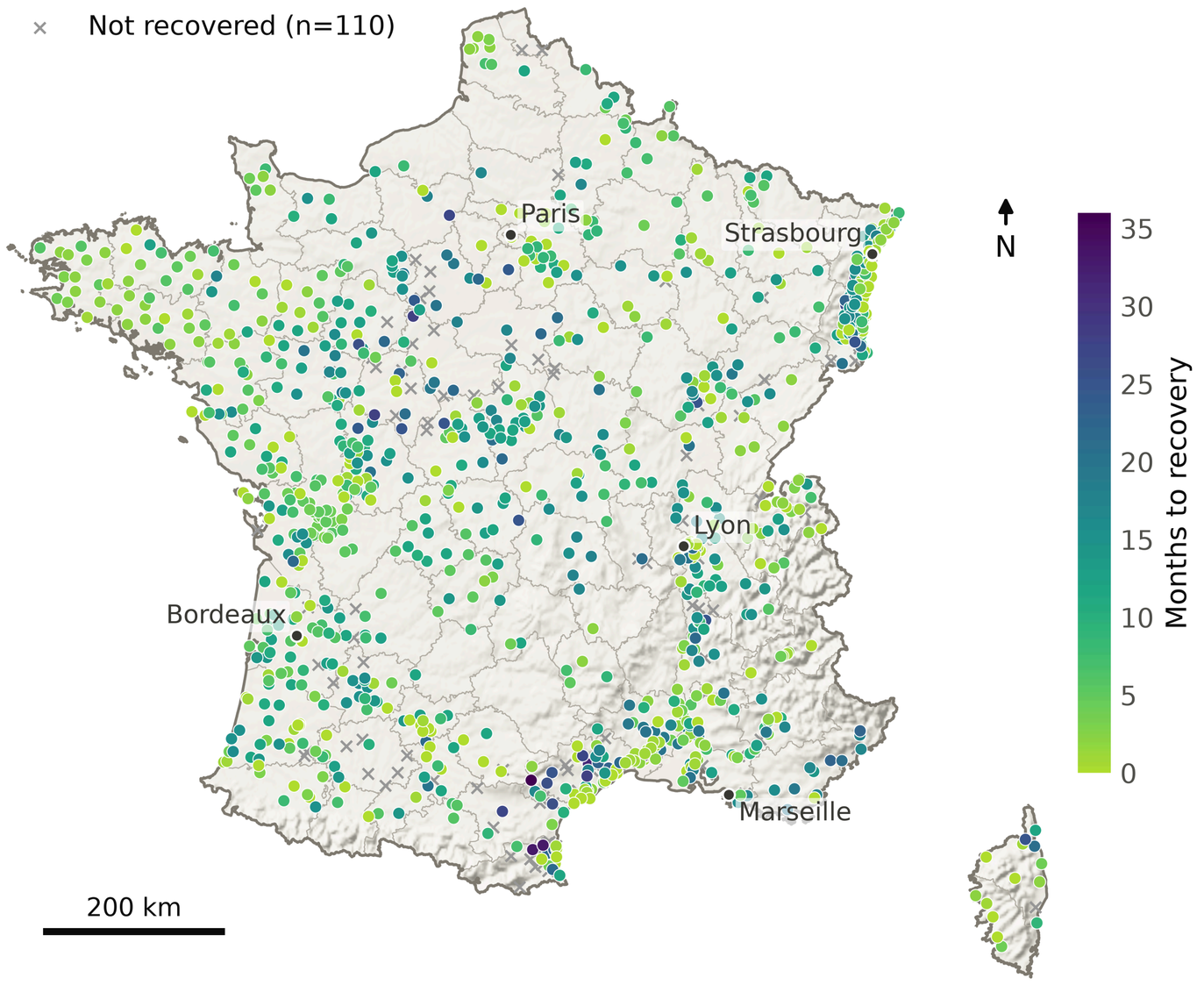


Fig 1. Months from each station’s drought minimum until the water table returned to its 2000–2019 monthly normal. Grey crosses mark the 110 stations still below normal in December 2025.

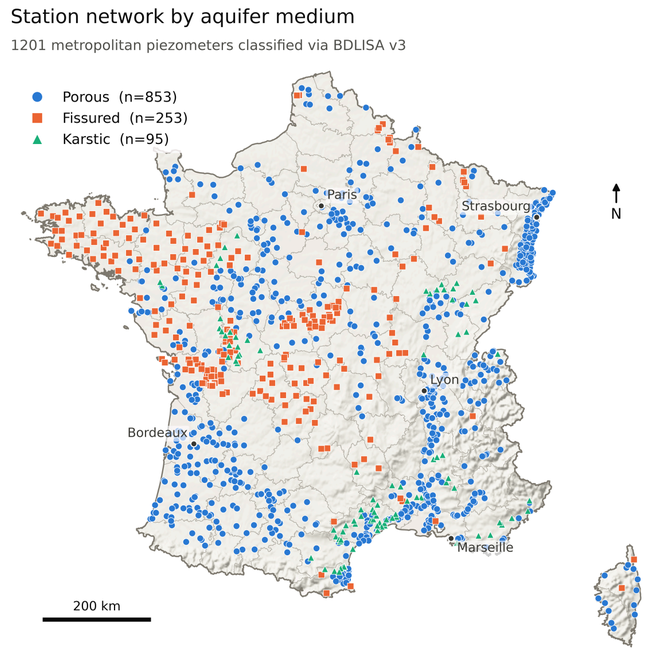


Fig 7. The 1,201 analysed stations by BDLISA medium. Fissured follows the crystalline basement, karstic the limestone massifs.

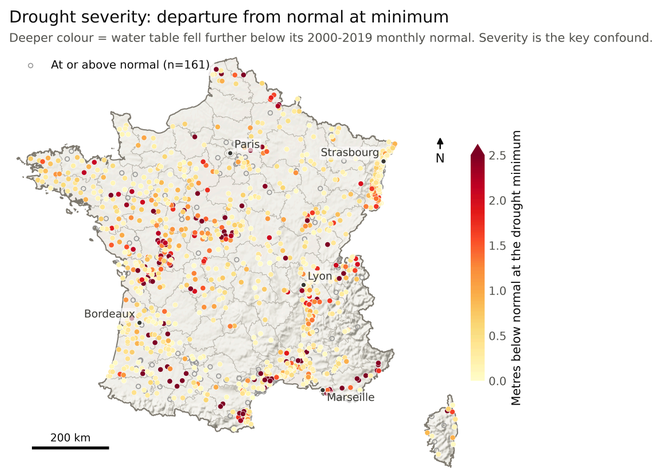


Fig 8. Departure from normal at the drought minimum. Severity clusters geographically, which is why it must be controlled for.

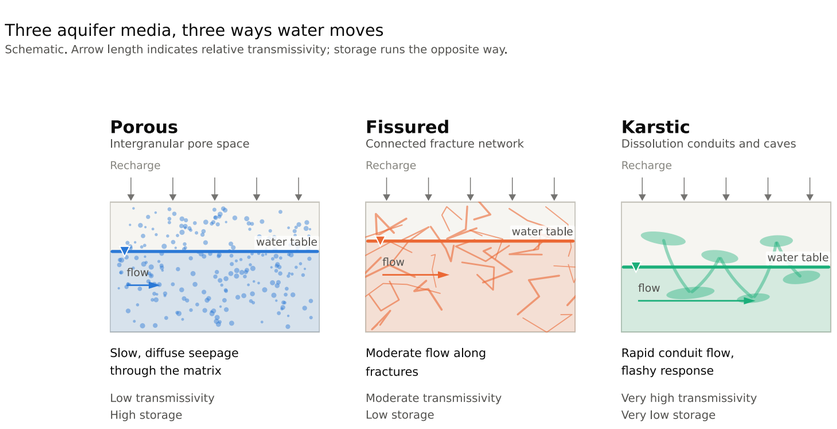


Fig 6. Why the media differ: pore space, fractures and conduits store and transmit water on different timescales.

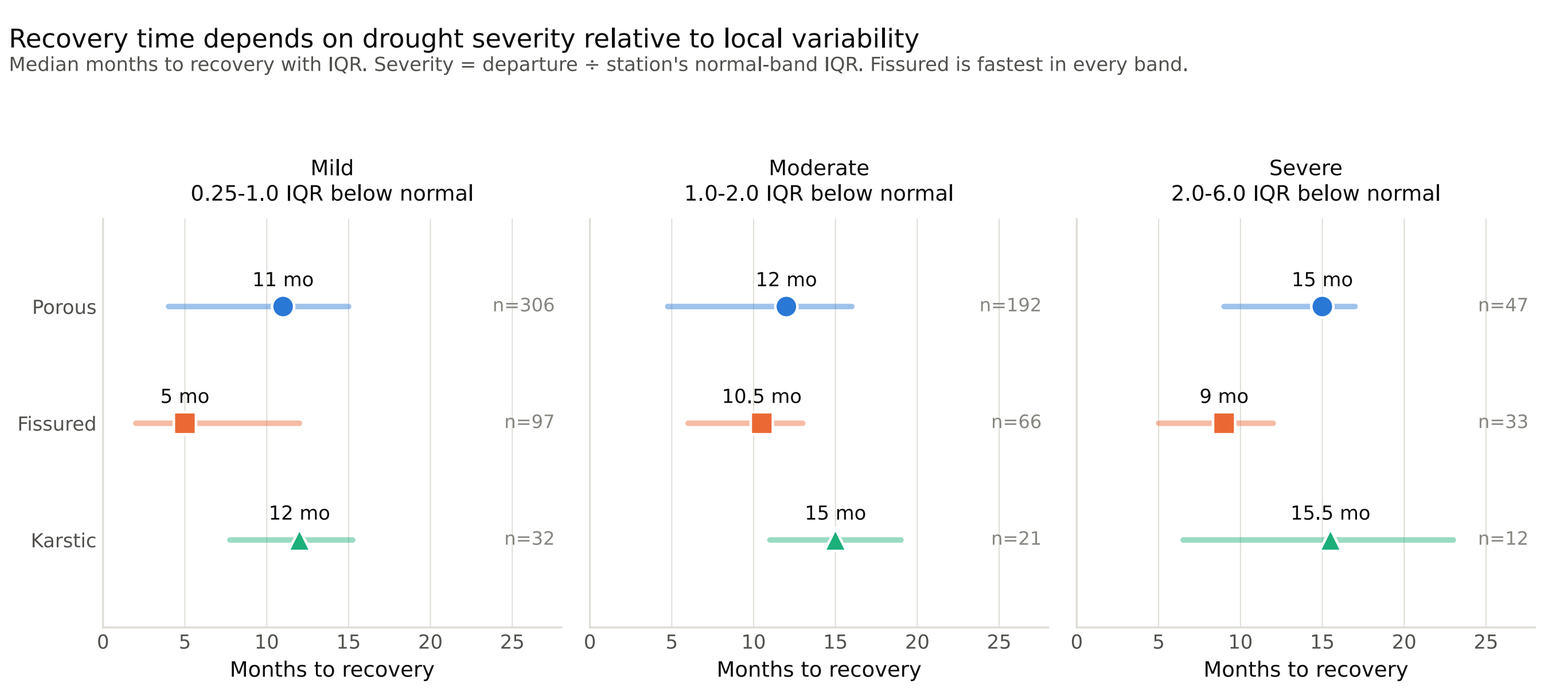


Fig 2. Recovery within bands of severity standardised by each station's own normal-band IQR. Fissured is fastest in all three bands; the ranking does not reverse.

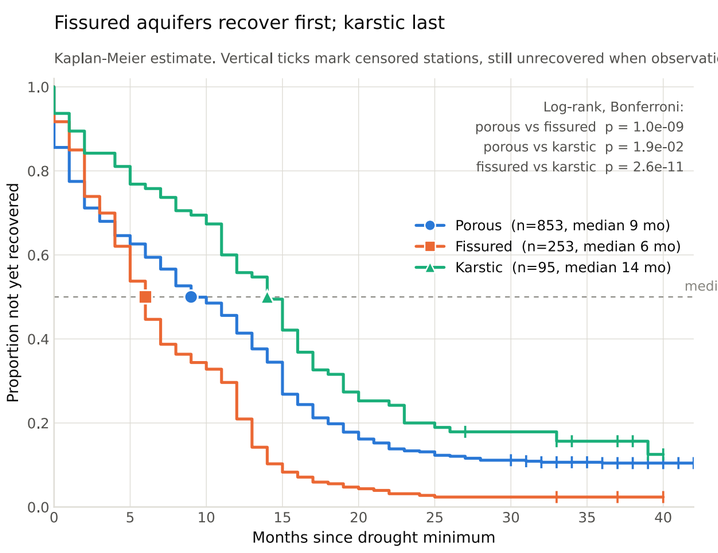


Fig 3. Kaplan–Meier estimate, which retains the censored stations. Log-rank separates all three media.

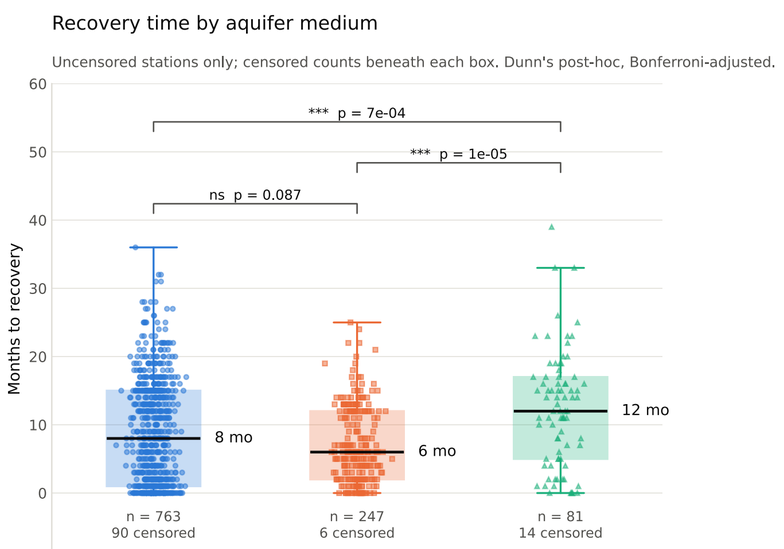


Fig 4. Pooled comparison (unadjusted for severity). Porous–fissured *ns* reflects mixing of severity levels; see Fig 2 for the stratified result and Fig 3 for the censoring-aware test.

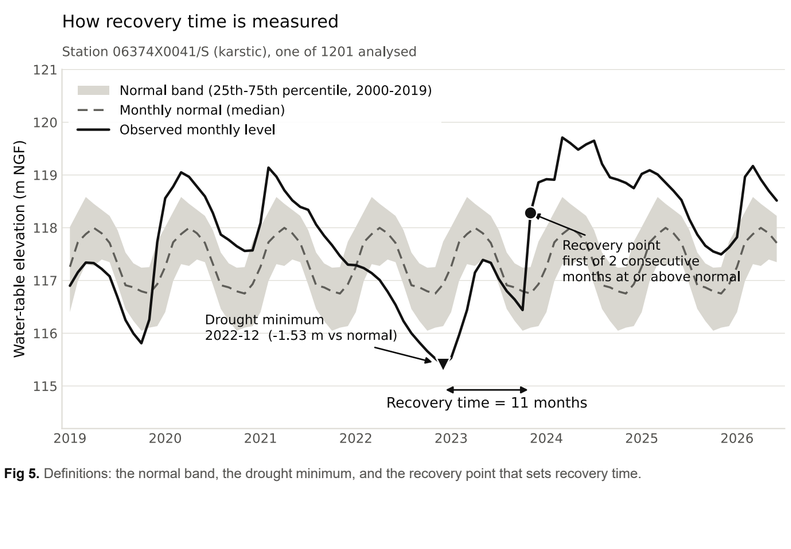


Fig 5. Definitions: the normal band, the drought minimum, and the recovery point that sets recovery time.

KEY FINDINGS

- **Fissured** aquifers recovered in **6 months** (median), **porous** in **8**, **karstic** in **12**.
- Fissured was fastest in **every severity stratum** ($p < 0.05$ in each); no ranking reversal once severity is standardised by station variability.
- **110 of 1,201 stations (9%)** had not recovered by December 2025 — disproportionately karstic and deep-drawdown.

LIMITATIONS

- Pumping influence not controlled; production wells may distort the recovery signal.
- A single drought event — results may not generalise.
- The severe karstic cell rests on n=12 uncensored of 20 (8 censored); that median is likely an underestimate.
- Association is not causation: medium correlates with climate, depth and thickness.
- BDLISA boundary generalisation; mixed-porosity entities (19%) excluded.

Data: Hub’Eau Piézométrie API (Licence Ouverte 2.0) · BDLISA v3 via Sandre (CC BY-NC-SA 2.0 FR) · Natural Earth (public domain)
Code and data: github.com/nellchris/aquifer-memory-map